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Enrollment number : A71004824054

Subject : cloud Computing

Q1. Define a Security Group and explain its role in EC2 networking.

Definition:
A **Security Group** in **Amazon EC2 (Elastic Compute Cloud)** is a **virtual firewall** that controls **inbound and outbound traffic** to and from your EC2 instances. It works at the **instance level** (not the subnet level) and helps ensure that only **authorized network traffic** can reach your cloud resources.

Each Security Group consists of a set of **rules** that define which traffic is allowed or denied based on:

- **Protocol** (e.g., TCP, UDP, ICMP)
- **Port number** (e.g., 22 for SSH, 80 for HTTP)
- **Source or destination IP address or CIDR block**

Role of Security Group in EC2 Networking:

1. **Controls Inbound Traffic (Ingress Rules):**
   Determines what kind of traffic is allowed **to enter** your EC2 instance.
   Example: Allow only SSH (port 22) access from your office IP.
2. **Controls Outbound Traffic (Egress Rules):**
   Defines what kind of traffic can **leave** your EC2 instance.
   Example: Allow your instance to access the internet (port 80 or 443).
3. **Acts as a Virtual Firewall:**
   It protects your EC2 instance by blocking unauthorized access and filtering traffic at the network interface level.
4. **Stateful Nature:**
   Security Groups are **stateful**, meaning if you allow incoming traffic on a port, the

response traffic is automatically allowed — you don't need to add an outbound rule for it.

5. **Instance-Level Protection:**
   Each EC2 instance can be associated with one or more Security Groups, giving you flexible control over its network security.
6. **Default Deny Rule:**
   By default, **all inbound traffic is denied**, and **all outbound traffic is allowed** until you modify the rules.

Example:

Suppose you launch a web server (EC2 instance) running a website.

You can configure the **Security Group** as follows:

- **Inbound Rules:**
  - Allow HTTP traffic on port 80 from anywhere (0.0.0.0/0)
  - Allow SSH access on port 22 only from your IP (203.0.113.5/32)
- **Outbound Rules:**
  - Allow all traffic so your instance can access updates or APIs on the internet.

This setup ensures your web server is publicly accessible while keeping SSH secure and restricted.

Q2. Perform the following tasks:

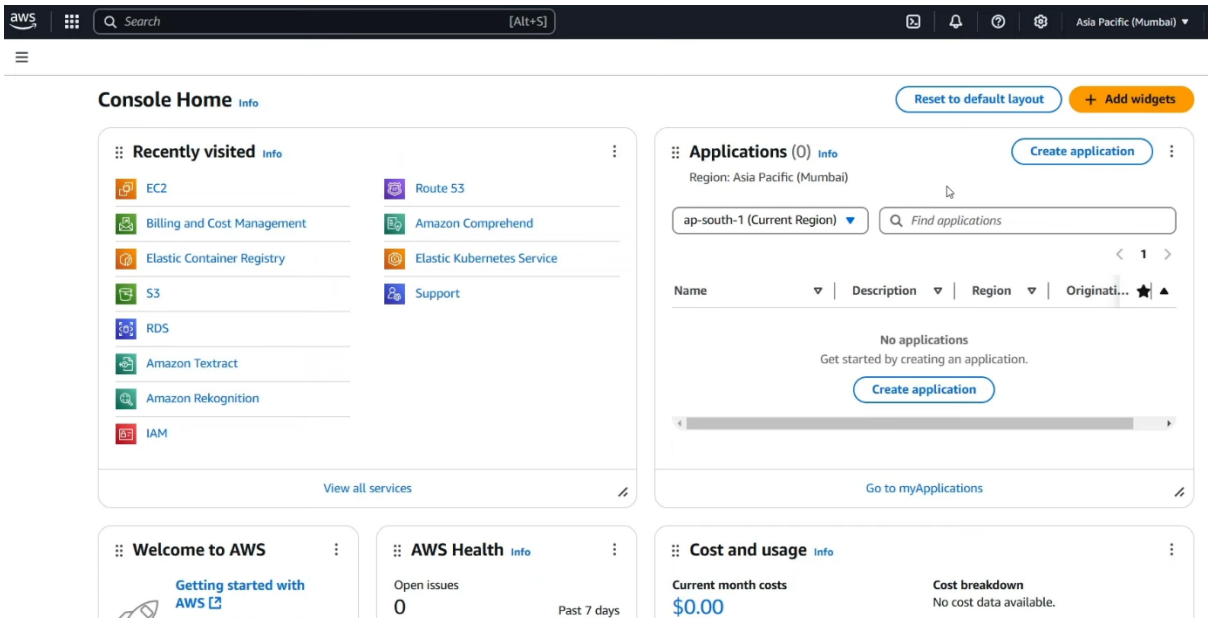
a) Create a new Security Group for an EC2 instance.

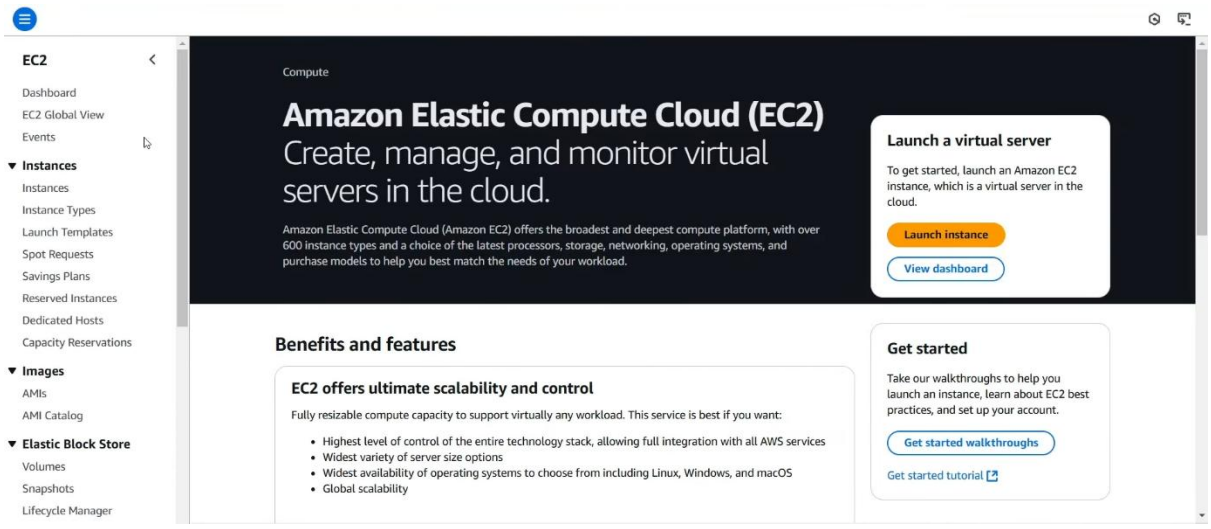
b) Configure inbound rules to allow SSH and HTTP access.

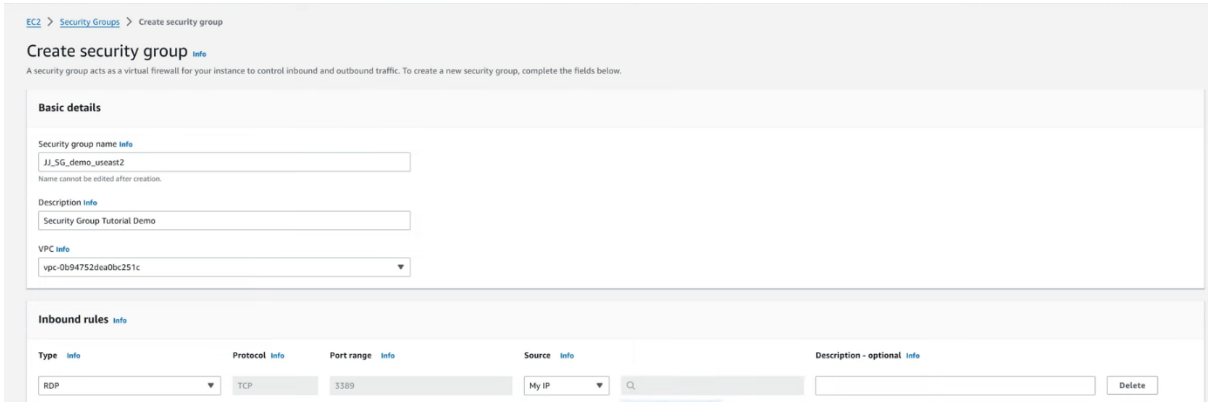
c) Launch an EC2 instance with the created Security Group.

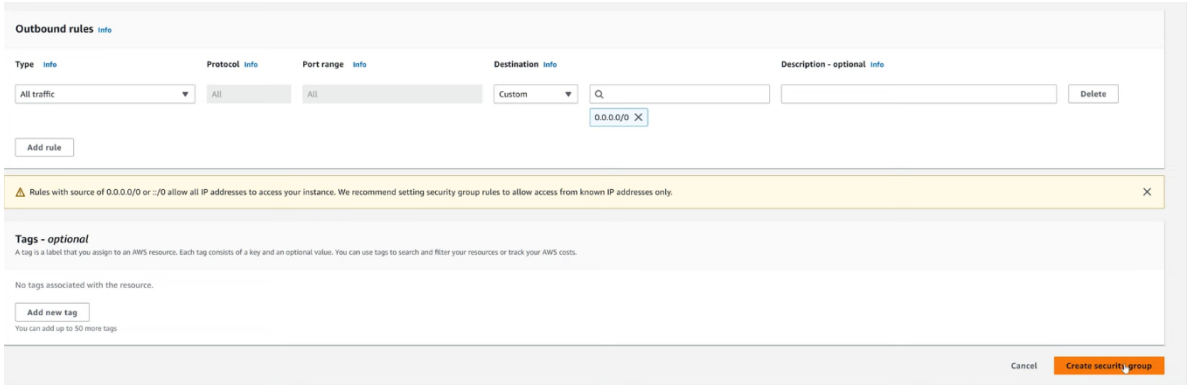
d) Test connectivity using SSH and web access.

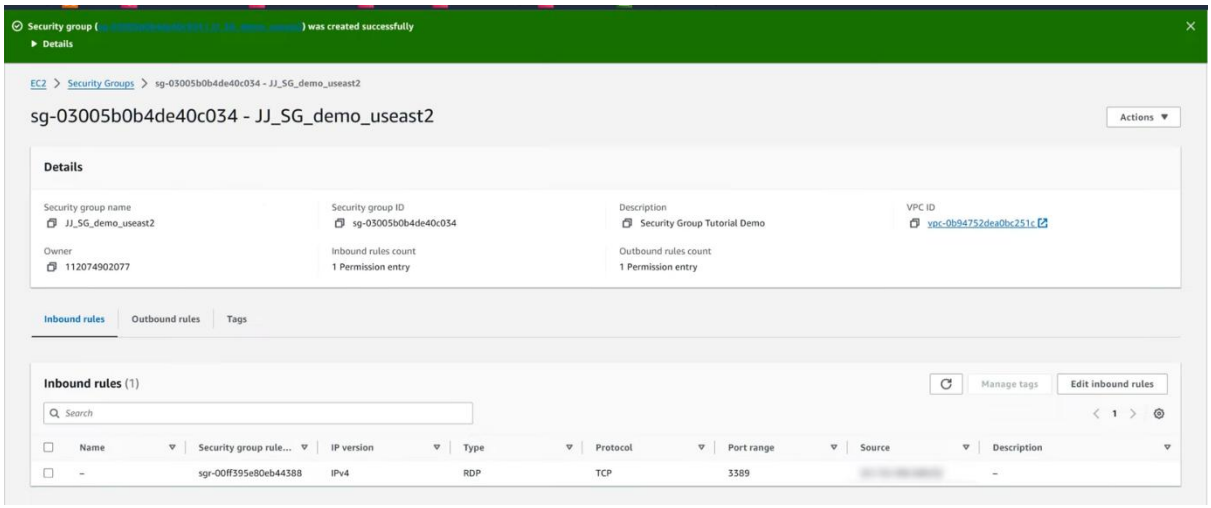
e) Modify Security Group rules to restrict access and observe changes.

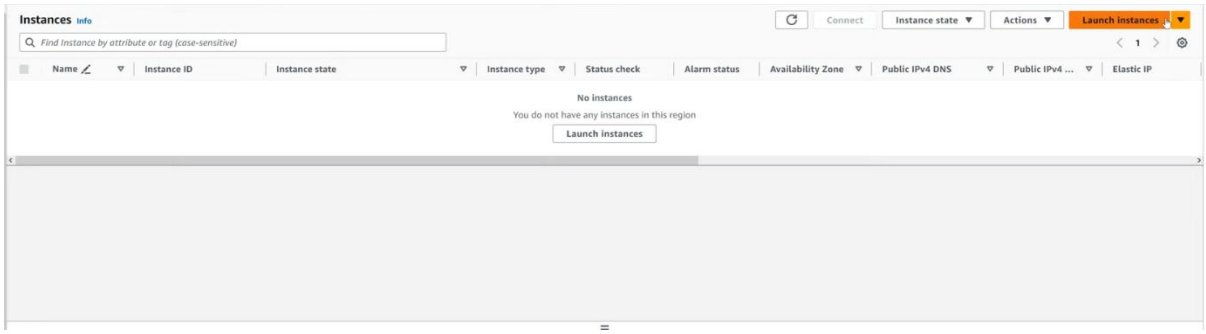

aws
Search
[Alt+S]
Asia Pacific (Mumbai)
Console Home Info
Reset to default layout
+ Add widgets
Recently visited Info
EC2
Billing and Cost Management
Elastic Container Registry
S3
RDS
Amazon Textract
Amazon Rekognition
IAM
Route 53
Amazon Comprehend
Elastic Kubernetes Service
Support
View all services
Applications (0) Info
Region: Asia Pacific (Mumbai)
Create application
ap-south-1 (Current Region)
Find applications
Name
Description
Region
Originati...
No applications
Get started by creating an application.
Create application
Go to myApplications
Welcome to AWS
Getting started with AWS
AWS Health Info
Open issues
0
Past 7 days
Cost and usage Info
Current month costs
$0.00
Cost breakdown
No cost data available.


EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Compute
Amazon Elastic Compute Cloud (EC2)
Create, manage, and monitor virtual servers in the cloud.
Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.
Launch a virtual server
To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.
Launch instance
View dashboard
Benefits and features
EC2 offers ultimate scalability and control
Fully resizable compute capacity to support virtually any workload. This service is best if you want:
Highest level of control of the entire technology stack, allowing full integration with all AWS services
Widest variety of server size options
Widest availability of operating systems to choose from including Linux, Windows, and macOS
Global scalability
Get started
Take our walkthroughs to help you launch an instance, learn about EC2 best practices, and set up your account.
Get started walkthroughs
Get started tutorial


EC2 > Security Groups > Create security group
Create security group Info
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.
Basic details
Security group name Info
JJ_SG_demo_useast2
Name cannot be edited after creation.
Description Info
Security Group Tutorial Demo
VPC Info
vpc-0b94752dea0bc251c
Inbound rules Info
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
RDP
TCP
3389
My IP
Delete


Outbound rules Info
Type Info
Protocol Info
Port range Info
Destination Info
Description - optional Info
All traffic
All
All
Custom
0.0.0.0/0
Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags
Cancel
Create security group


Security group ( ) was created successfully
Details
EC2 > Security Groups > sg-03005b0b4de40c034 - JJ_SG_demo_useast2
sg-03005b0b4de40c034 - JJ_SG_demo_useast2
Actions
Details
Security group name
JJ_SG_demo_useast2
Security group ID
sg-03005b0b4de40c034
Description
Security Group Tutorial Demo
VPC ID
vpc-0b94752dea0bc251c
Owner
112074902077
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule...
IP version
Type
Protocol
Port range
Source
Description
-
sgr-00ff395e80eb44388
IPv4
RDP
TCP
3389
-


Instances Info
Connect
Instance state
Actions
Launch instances
Find instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
No instances
You do not have any instances in this region
Launch instances

EC2 > Instances > Launch an instance

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

EC2 Demo Instance

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Li

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI — Free tier eligible

ami-0cd3c7f72edd5b06d (64-bit (x86), uefi-preferred) / ami-0beec7f2863e681cf (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.3.2...read more

ami-0cd3c7f72edd5b06d

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel | Launch instance | Review commands

required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Selected AMI: (ami-0cd3c7f72edd5b06d) (Quickstart AMIs)

Search for an AMI by entering a search term e.g. "Windows"

Quickstart AMIs (47) — Commonly used AMIs | My AMIs (0) — Created by me | AWS Marketplace AMIs (8976) — AWS & trusted third-party AMIs | Community AMIs (500) — Published by anyone

Refine results

Clear all filters

Free tier only Info

OS category

All Linux/Unix

All Windows

Architecture

64-bit (Arm)

32-bit (x86)

64-bit (x86)

64-bit (Mac)

64-bit (Mac-Arm)

All products (6 filtered, 47 unfiltered)

Microsoft Windows Server 2022 Base
ami-0b24eb32c43d56c23 (64-bit (x86))
Microsoft Windows 2022 Datacenter edition. [English]
Platform: windows | Root device type: ebs | Virtualization: hvm | ENA enabled: Yes
Free tier eligible | Verified provider | Select | 64-bit (x86)

Microsoft Windows Server 2022 Core Base
ami-002885018 2b686e1f (64-bit (x86))
Microsoft Windows 2022 Datacenter Core edition. [English]
Platform: windows | Root device type: ebs | Virtualization: hvm | ENA enabled: Yes
Free tier eligible | Verified provider | Select | 64-bit (x86)

Microsoft Windows Server 2019 Base
ami-06fe4639440b3ab22 (64-bit (x86))
Microsoft Windows 2019 Datacenter edition. [English]
Platform: windows | Root device type: ebs | Virtualization: hvm | ENA enabled: Yes
Free tier eligible | Verified provider | Select | 64-bit (x86)

Microsoft Windows Server 2016 Base
ami-0dd478adda4cc704d (64-bit (x86))
Microsoft Windows 2016 Datacenter edition. [English]
Free tier eligible | Select | 64-bit (x86)

On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand RHEL base pricing: 0.0716 USD per Hour

Additional costs apply for AMIs with pre-installed software

Compare instance types

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

JJ-Demo-TodaysDate

Create new key pair

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

Network settings Info

Edit

Network Info

vpc-0b94752dea0bc251c

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

We'll create a new security group called **'launch-wizard-2'** with the following rules:

- [x] Allow RDP traffic from — Anywhere 0.0.0.0/0
  Helps you connect to your instance
- [ ] Allow HTTPS traffic from the internet
  To set up an endpoint, for example when creating a web server
- [ ] Allow HTTP traffic from the internet

Summary

Number of instances Info

1

Software Image (AMI)
Microsoft Windows Server 2022 ...read more
ami-0b24eb32c43d56c23

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel | Launch instance | Review commands

EC2 > Instances > i-0e3a32d2681e90f15

Instance summary for i-0e3a32d2681e90f15 (EC2 Demo Instance)

Updated less than a minute ago

Connect | Instance state | Actions

Remote Desktop Connection

Instance ID: i-0e3a32d2681e90f15 (EC2 Demo Instance)		Private IPv4 addresses: 172.31.39.74
IPv6 address: –		Public IPv4 DNS: ec2-3-141-153-221.us-east-2.compute.amazonaws.com |open address
Hostname type: IP name: ip-172-31-39-74.us-east-2.compute.internal		
Answer private resource DNS name: IPv4 (A)	Instance type: t2.micro	Elastic IP addresses: –
Auto-assigned IP address: 3.141.153.221 [Public IP]	VPC ID: vpc-0b94752dea0bc251c	AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM Role: –	Subnet ID: subnet-00619de92d8030b3d	Auto Scaling Group name: –
IMDSv2: Optional. EC2 recommends setting IMDSv2 to required | Learn more		

Details | Security | Networking | Storage | Status checks | Monitoring | Tags

Instance details Info

Platform: windows	AMI ID: ami-0b24eb32c43d56c23	Monitoring: disabled
Platform details: Windows	AMI name: Windows_Server-2022-English-Full-Base-2023.12.13	Termination protection: Disabled
Stop protection: Disabled	Launch time: Tue Jan 09 2024 10:53:26 GMT-0600 (Central Standard Time) (4 minutes)	AMI location: amazon/Windows_Server-2022-English-Full-Base-2023.12.13
Instance auto-recovery: Default	Lifecycle: normal	Stop-hibernate behavior: Disabled
AMI Launch index: 0	Key pair assigned at launch: JJ-Demo-TodaysDate	State transition reason: –

EC2 > Instances > i-0e3a32d2681e90f15 > Connect to instance

Connect to instance Info

Connect to your instance i-0e3a32d2681e90f15 (EC2 Demo Instance) using any of these options

Session Manager | RDP client | EC2 serial console

Instance ID

i-0e3a32d2681e90f15 (EC2 Demo Instance)

Connection Type

Connect using RDP client
Download a file to use with your RDP client and retrieve your password.

Connect using Fleet Manager
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see Working with SSM Agent

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file

When prompted, connect to your instance using the following details:

Public DNS
ec2-3-141-153-221.us-east-2.compute.amazonaws.com

Username
Administrator

Password Get password

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

Cancel

Instances Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states | 1

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4

No instances
You do not have any instances in this region
Launch instances

EC2 > Instances > Launch an instance

It seeems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Do not show me this message again

Take a walkthrough

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

e.g. My Web Server

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

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Browse more AMIs

Including AMIs from AWS, Marketplace and

Summary

Number of instances Info

1

Software Image (AMI)

-

Virtual server type (instance type)

t2.micro

Firewall (security group)

-

Storage (volumes)

-

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage,